Hierarchical Clustering based Multiple Particle Filtering for Data Assimilation

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***Abstract*—Data assimilation (DA) combines system models with noisy observations to enhance state estimation. Due to the highly nonlinear nature of DA and the curse of dimensionality in particle filtering (PF), the multiple particle filter (MPF) has been proposed to partition the state space into several subspaces so that a PF can be run independently on each subspace. However, while state partitioning in MPF reduces the variance of state estimates, it also introduces un additional bias due to the loss of cross-correlations between subspaces. Consequently, the performance of MPF heavily depends on the design of the state partitioning scheme, as it involves a bias-variance trade-off.**

In this paper, we propose a hierarchical clustering-based state partitioning scheme for MPF (HC-MPF). This method groups highly correlated state variables within the same subspace while assigning weakly correlated ones to separate subspaces. HC-MPF not only enhances the performance of MPF but also offers a structured framework for analyzing complex dynamical systems. Numerical experiments demonstrate that HC-MPF outperforms existing partitioning techniques in nonlinear DA applications.

***Index Terms*—Multiple particle filter, state space partitioning, hierarchical clustering, data assimilation.**

I. Introduction

Data assimilation (DA) [1]–[3] is the discipline that combines system models with noisy observations to enhance state estimation. It plays a crucial role in various fields, including weather forecasting [4], oceanography [5], glaciology [6], medicine and biology [7], as well as the human and social sciences [8]. Within the Bayesian framework, the Kalman filter (KF) [9] provides optimal state estimations when the DA models are linear and Gaussian. However, due to the highly nonlinear nature of most DA models, the extended Kalman filter (EKF) [10], the ensemble Kalman filter (EnKF) [11] and the particle filter (PF) [12] are widely used. Moreover, DA models are often highly complex in practical applications and their performance degrades significantly in high dimensional spaces due to the curse of dimensionality [13] [14].

Over the past decades, various methods have been developed to address this limitation, including high-dimensional distribution proposals [15] [16], advanced resampling strategies [17], hybrid methods [18], and localization techniques [19]–[22]. The multiple particle filter (MPF) [22] is one of the powerful methods. It partitions the state space into serval disjoint subspaces of lower dimension and runs a PF independently on each subspace. Each PF approximates a marginal posterior using a set of weighted samples. To exchange information between subspaces, the estimated values at the previous time step and the predicted values at the current time step of the state variables in other subspaces must be transmitted to each PF for particle propagation during the prediction stage and for weight calculation in the update step.

However, while state partitioning in the MPF reduces the variance of state estimates, it also introduces an additional bias due to the loss of cross-correlations between subspaces. Therefore, the performance of the MPF heavily depends on the design of the state partition scheme, as it involves a bias-variance trade-off. Some state space partitioning schemes have been proposed to enhance the performance of MPF. The work in [23] proposes an adaptive state partitioning scheme for MPF in the multiple-object tracking problem. Based on the second order statistics of each PF, a PF may be split into several smaller PFs in the next time step depending on the cross-correlation between objects. Conversely, two PFs may be merged in the next time step based on their KL divergence. However, this method is not applicable to DA problems. Another approach in [24] introduces a topology based partitioning scheme that relies on the connectedness of state variables over time. However, it only accounts for correlations among state variables in the transition matrix, assuming that the topology of the transition equation is known. The effects of process noise on state correlations, as well as the correlations of state variables in nonlinear and/or non-Gaussian models, are not considered. This assumption is relatively uncommon in practice.

The advent of machine learning, particularly unsupervised learning techniques [25] [26], has opened novel possibilities for designing state space partitioning schemes that can fully exploit correlations among state variables. In this paper, we propose a hierarchical clustering based state space partitioning scheme for MPF (HC-MPF). Inspired by the EnKF, the state correlation matrix is estimated using the Jacobian matrix of the transition equation. Based on this matrix, our HC-MPF groups the most correlated state variables within the same subspace while assigning the least correlated ones to separate subspaces. This approach not only enhances the performance of MPF but also provides a framework for analyzing the structure of complex dynamical systems, including nonlinear systems.

The paper is organized as follows. Section II provides a brief overview of the data assimilation problem and the multiple particle filter. Section III details the methodology for generating a hierarchical clustering-based state-space partition for the multiple particle filter. Numerical results are presented in Section IV and conclusions are given in Section V.

II. Background

A. Data assimilation problem

Considering the classical data assimilation (DA) problem formulated within the framework of the state space model (SSM):

$$\mathbf{x}_t = f_t(\mathbf{x}_{t-1}, \mathbf{w}_t) \tag{1}$$

$$\mathbf{y}_t = h_t(\mathbf{x}_t, \mathbf{v}_t) \tag{2}$$

where

- $\mathbf{x}_t \in \mathbb{R}^{d_x}$ denotes the hidden state of the system that requires to be estimated at each time step, and $\mathbf{y}_t \in \mathbb{R}^{d_y}$ represents the corresponding observation at time t,
- the process noise $\mathbf{w}_t$ accounts for the stochastic uncertainties in the state transition, and the observation noise $\mathbf{v}_t$ represents perturbations in the observed data,
- the functions $f(\cdot)$ and $h(\cdot)$ are respectively the transition equation and the observation equation, which can be nonlinear.

The goal of data assimilation is to estimate the current state $\mathbf{x}_t$ by integrating the models $f_t(\cdot)$ and $h_t(\cdot)$ with all available observations $\mathbf{y}_1, \mathbf{y}_2, \cdots, \mathbf{y}_t$, thereby improving the accuracy of predictions in dynamical systems.

B. Multiple Particle Filter

Due to the highly non-linear nature of DA and the curse of dimensionality in PF, the MPF has been proposed to partition the state space into several subspaces so that a PF can be run independently on each subsapce. Within this framework, each PF approximates the marginal posterior distribution $p(\mathbf{x}_{k,t}|\mathbf{y}_t)$ through a set of samples $\{\mathbf{x}_{k,t}^{(i)}\}_{i=1}^{N_p}$ and their associated weights $\{w_{k,t}^{(i)}\}_{i=1}^{N_p}$ on k^{th} subspace. The workflow of the MPF at each time step can be summarized in the following steps:

1. First, the state space $\mathbf{x}_t$ is divided into K subspaces according to a specified partitioning scheme, i.e. $\mathbf{x}_t^T = [\mathbf{x}_{1,t}^T, \mathbf{x}_{2,t}^T, \cdots, \mathbf{x}_{K,t}^T]$.
2. Next, the particles in the kth subspace $\mathbf{x}_{k,t}^{(i)}$ are propagated as:
$$\mathbf{x}_{k,t}^{(i)} \sim p(\mathbf{x}_{k,t}|\mathbf{x}_{k,t-1}^{(i)}, \hat{\mathbf{x}}_{-k,t-1}), \quad i = 1, \cdots, N_p \tag{3}$$
where $\hat{\mathbf{x}}_{-k,t-1}$ represents the estimated state $\hat{\mathbf{x}}_{t-1}$ from the previous time step excluding the kth subspace, i.e. $\hat{\mathbf{x}}_{-k,t-1} = [\hat{\mathbf{x}}_{1,t-1}^T, \cdots, \hat{\mathbf{x}}_{k-1,t-1}^T, \hat{\mathbf{x}}_{k+1,t-1}^T, \cdots, \hat{\mathbf{x}}_{K,t-1}^T]$.
3. Then, the local weights $w_{k,t}^{(i)}$ are updated by:
$$w_{k,t}^{(i)} \propto \frac{p(\mathbf{y}_t|\mathbf{x}_{k,t}^{(i)}, \tilde{\mathbf{x}}_{-k,t})p(\mathbf{x}_{k,t}^{(i)}|\hat{\mathbf{x}}_{k,t-1})}{q(\mathbf{x}_{k,t}^{(i)}|\hat{\mathbf{x}}_{k,t-1}, \mathbf{y}_t)}, \quad \text{with} \sum_{i=1}^{N_p} w_{k,t}^{(i)} = 1 \tag{4}$$
where $\tilde{\mathbf{x}}_{k,t}$ is the predicted state in the kth subspace, i.e. $\tilde{\mathbf{x}}_{k,t} = \sum_{i=1}^{N_p} w_{k,t-1}^{(i)} \mathbf{x}_{k,t}^{(i)}$.
4. Finally, the resampling step is applied in each subspace.

The use of a larger number of subspaces in the MPF has been shown to significantly reduce the variance of the estimates. However, this also results in a greater loss of cross-correlation between subspaces and an additional bias. Therefore, a bias-variance trade-off must be considered when designing the state space partitioning for the MPF. The performance of the MPF is highly dependent on the selected state space partitioning scheme.

III. Hierarchical clustering based state space partitioning

The advent of unsupervised learning techniques, most notably hierarchical clustering [27], has engendered novel possibilities for designing state space partitioning schemes that can fully exploit correlations among state variables. In this paper, we propose the use of hierarchical clustering for the state space partitioning using a nonsimilarity matrix. Initially, each state variable is treated as an individual cluster (root node), and is then progressively merged with the most similar clusters. This hierarchical process continues until a single cluster (a tree-like structure) is formed. Therefore, we focus on the estimation of the state vector's non-similarity matrix.

Considering that the particles in the kth subspace $\mathbf{x}_{k,t}^{(i)}$ are propagated as $\mathbf{x}_{k,t}^{(i)} \sim p(\mathbf{x}_{k,t}|\mathbf{x}_{k,t-1}^{(i)}, \hat{\mathbf{x}}_{-k,t-1})$, the cross-correlations between the subspaces are not taken into account in the MGF. Therefore, it is essential to group the most correlated state variables within the same subspaces, while the least correlated variables are placed in different subspaces. The hierarchical clustering (HC) based state space partitioning scheme using a non-similarity matrix at the current time step t can be summarized in the following steps:

1. After the resampling step at time step $t-1$, the covariance matrix can be obtained from the N_p uniformly distributed particles based on the sample covariance matrix:
$$\hat{\mathbf{P}}_{t-1|t-1} = \frac{1}{N_p - 1} \sum_{i=1}^{N_p} \left(\mathbf{x}_t^{(i)} - \bar{\mathbf{x}}_t\right) \left(\mathbf{x}_t^{(i)} - \bar{\mathbf{x}}_t\right)^\top \tag{5}$$
where:
$$\bar{\mathbf{x}}_t = \frac{1}{N_p} \sum_{i=1}^{N_p} \mathbf{x}_t^{(i)}. \tag{6}$$
2. Due to the non linearity nature of DA, the covariance matrix $\hat{\mathbf{P}}_{t|t-1}$ of the predicted posterior $p(\mathbf{x}_t|\mathbf{y}_{1:t})$ is estimated as:
$$\hat{\mathbf{P}}_{t|t-1} = \mathbf{J}_m \hat{\mathbf{P}}_{t|t-1} \mathbf{J}_m^T + \mathbf{Q}_t \tag{7}$$
where $\mathbf{J}_m \in \mathbb{R}^{d_x \times d_x}$ is the Jacobian matrix, which is obtained by linearizing the state transition function $f(\cdot)$ at the estimated state from the previous time step $t-1$: $\mathbf{J}_m = \left.\frac{\partial f(\mathbf{x})}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_{t-1}}$.

3. Accoding to this covariance matrix, the element $\hat{c}_{ij}$ state correlation matrix $\hat{\mathbf{C}}_{t|t-1}$ is obtained by:

$$\hat{c}_{ij} = \begin{cases} 1, & \text{if } i = j, \\ \frac{\hat{p}_{ij}}{\sqrt{\hat{p}_{ii}\hat{p}_{jj}}}, & \text{if } i \neq j. \end{cases} \tag{8}$$

To eliminate some disturbances, a threshold τ is set as $\tau = 0.3$:

$$\hat{c}_{ij} = \begin{cases} \hat{c}_{ij}, & \text{if } |\hat{c}_{ij}| > \tau, \\ 0, & \text{else.} \end{cases} \tag{9}$$

4. The non-similarity matrix of HC is defined by the distance matrix $\mathbf{D}_{t|t-1}$, which is given by:

$$\mathbf{D}_{t|t-1} = \mathbf{1}_{d_x \times d_x} - |\hat{\mathbf{C}}_{t|t-1}| \tag{10}$$

5. The HC method is applied to obtain the result of state partitioning into K subspaces.

Once the hierarchical clustering based state space partitioning scheme is obtained, it will be assigned in SectionII-B.1.

IV. Simultion results

This section provides the experimental validation of the proposed state space partitioning method for the MPF. The experiments are conducted in the framework of a classical DA problem and highlight the benefits of the hierarchical clustering-based state space partitioning scheme. The Lorenz 96 model is a low-dimensional chaotic system widely used in data assimilation for weather predictions and climate simulations studies. It represents the evolution of some unspecified scalar meteorological quantity (perhaps vorticity or temperature) over a latitude circle [28]. The temporal dynamics of this model are characterized by a set of ordinary differential equations (ODEs):

$$\begin{aligned} \frac{\mathrm{d}x_t(n)}{\mathrm{d}t} &= [x_t(n+1) - x_t(n-2)]\, x_t(n-1) \\ &\quad - x_t(n) + F, \quad \forall n \in [1, d_x] \end{aligned} \tag{11}$$

where $x_t(n)$ denotes the nth state variable of $\mathbf{x}_t$ and the indices follow periodic boundary conditions: $x_t(-1) = x_t(d_x - 1)$, $x_t(0) = x_t(d_x)$ and $x_t(d_x + 1) = x_t(1)$. These ODEs are numerically integrated using a fourth-order Runge–Kutta method with a time step of a 0.05 time unit. In the standard configuration, the system dimension is set to $d_x = 40$ and the forcing constant is $F = 8$. The initial state vector is $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, 0.01 \times \mathbf{I})$. The observation system $\mathbf{y}_t$ is defined as:

$$\mathbf{y}_t = \mathbf{x}_t + \mathbf{v}_t, \quad \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{12}$$

In experiments, we compare two versions of the MPF: random partitioning (RM-BPF) in [24] and hierarchical clustering based partitioning (HC-MPF). In RM-MPF, the state variables are randomly assigned to K subspaces at each time step, which is the simplest partitioning scheme. The size of the subspaces in this case can be adjusted based on specific requirements. Performances are given in terms of mean square error (MSE), bias and variance which are obtained over $N_s = 100$ simulations of length $T = 100$ time steps. The MSE is given by:

$$\text{MSE} = \frac{1}{d_x}\frac{1}{T}\sum_{t=1}^{T}\frac{1}{N_s}\sum_{s=1}^{N_s} \parallel \hat{\mathbf{x}}_t^s - \mathbf{x}_t \parallel_2^2 \tag{13}$$

where $\|.\|_2$ is the L_2 norm in $\mathbb{R}^{d_x}$. The bias is obtained by:

$$\text{Bias} = \frac{1}{d_x}\frac{1}{T}\sum_{t=1}^{T} \parallel \bar{\mathbf{x}}_t^s - \mathbf{x}_t \parallel_1 \tag{14}$$

where $\|.\|_1$ is the L_1 norm in $\mathbb{R}^{d_x}$ and $\bar{\mathbf{x}}_t^s = \frac{1}{N_s}\sum_{s=1}^{N_s}\hat{\mathbf{x}}_t^s$. The variance is given by:

$$\text{Variance} = \frac{1}{d_x}\frac{1}{T}\sum_{t=1}^{T} \parallel \frac{1}{N_s}\sum_{s=1}^{N_s}\left(\hat{\mathbf{x}}_t^s - \bar{\mathbf{x}}_t^s\right)^2 \parallel_1 \tag{15}$$

A. Influence of the number of blocks

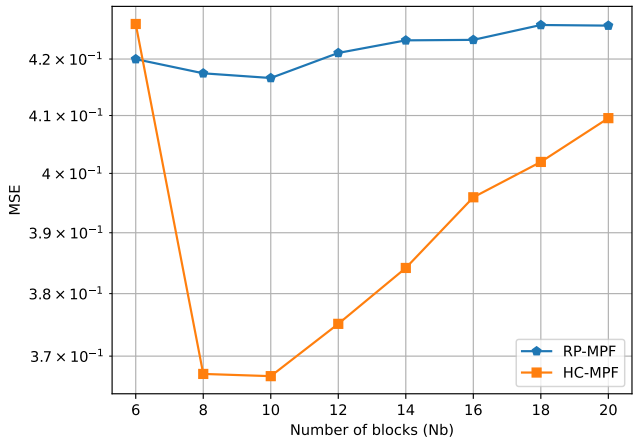
In this section, $N_p = 1000$ particles are not sufficient for certain numbers of subspaces. The curse of dimensionality must be taken into account. Therefore, the sizes of the subspaces in RM-MPF are different. Fig.1 shows the performance of state estimation in terms of MSE, bias and variance versus the number of subspaces.

As illustrated in Fig. 1a, our HF-MPF significantly outperforms the RM-MPF when the number of subspaces $K > 6$. This indicates that HC-MPF more effectively groups correlated state variables within the same subspace. When $K = 6$, each subspace remains relatively large. Consequently, $N_p = 1000$ particles is insufficient for the filters on each subspace and MPFs with both partitioning schemes suffer from the curse of dimensionality. As the number of subspaces K increases, the performances of both MPFs improve up to $K = 10$ and then degrade. Figure 1b shows that the bias of our HF-MPF remains consistently lower than that of RM-MPF. This suggests that the cross-correlation between subspaces with HC-MPF is relatively small. As the number of subspaces K increases, the bias of both MPFs increases due to a greater loss of cross-correlation between the subspaces. From Fig. 1c, the variances of both MPFs decrease rapidly as the number of subspaces K increases. As a consequence, a bias-variance trade-off is observed for both MPFs with $K = 10$ subspaces for $N_p = 1000$ particles, where both the bias and variance are relatively small.

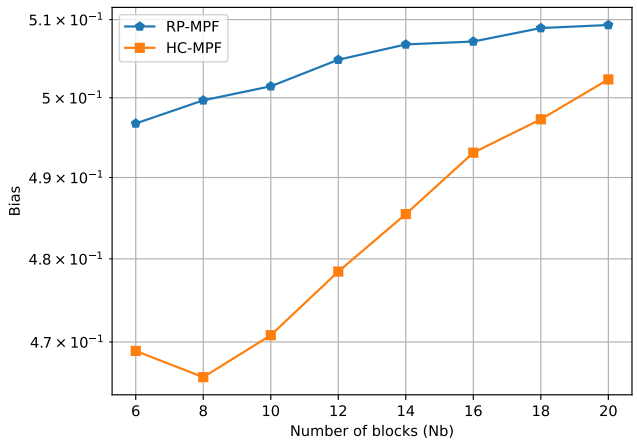
B. Influence of the number of particles

Considering the loss of cross-correlation between subspaces in MPF, we selected 10 subspaces for the simulation in this subsection. In this case, the sizes of subspaces in both MPFs are relatively small, and $N_p \geq 200$ particles is sufficient. Therefore, the sizes of subspaces in RM-MPF are the same.

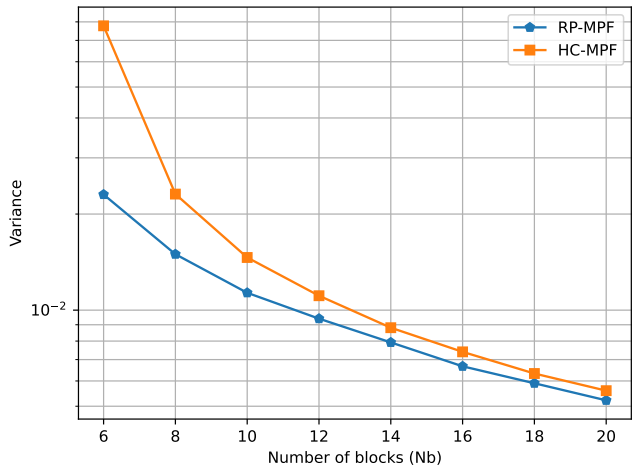
Fig. 2 shows the MSE as a function of the number of particles N_p. As the number of particles increases, the performance of both MPFs improves due to the mitigation of the curse of dimensionality and the estimated state correlation matrix $\hat{\mathbf{C}}_{t|t-1}$ in HC-MPF becomes more precise. This provides HC-MPF with more opportunities to group correlated state variables



(a) MSE



(b) Bias



(c) Variance

Fig. 1: Performance vs the number of subspaces K for $N_p = 1000$ particles

within the same subspaces. Therefore, for any given number of particles N_p, our HC-MPF significantly outperforms RM-MPF, even though certain subspace sizes in HC-MPF may be larger than those in RM-MPF for some particle numbers.

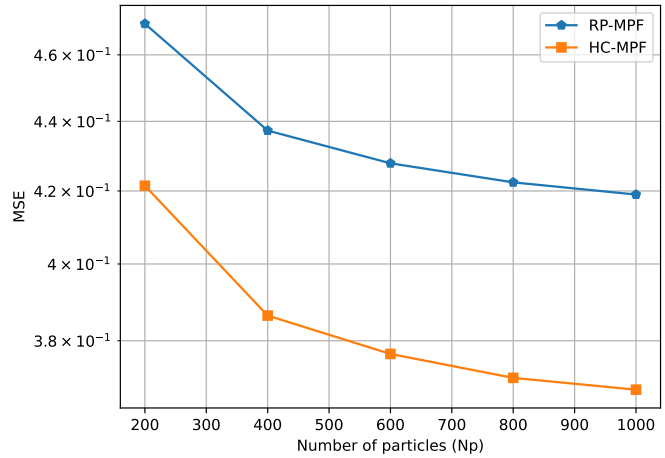


Fig. 2: Performance vs the number of particles N_p for 10 subsapces

V. Conclusion

The use of state partitioning in the multiple particle filter reduces the variance of state estimates. However, it also introduces an additional bias due to the loss of cross-correlations between subspaces. Consequently, the performance of MPF heavily depends on the design of the state partitioning scheme, as it involves a bias-variance trade-off. In this paper, we proposed a hierarchical clustering based state space partitioning scheme for MPF. By leveraging the Jacobian matrix of the transition equation to estimate state correlation matrix, our HF-MPF automatically groups highly correlated state variables within the same subspace while separating weakly correlated ones. This method not only enhances the performance of MPF by mitigating the loss of cross-correlations but also provides a systematic framework for analyzing the structure of complex dynamical systems. Numerical experiments demonstrate that our HC-MPF outperforms existing methods in nonlinear and high-dimensional data assimilation problems. In future work, we will focus on determining the optimal number of subspaces given a fixed number of particles based on the bias-variance trade-off in the performance of MPF.

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